Understanding Simple Water Bodies Mapping (SWBM) Script in EO Browser

Script: Simple Water Bodies Mapping (SWBM) | Source: Link
The script can be applied in sentinel-2 Level1C, Sentinel-2 Level2A, and Landsat datasets for the pixel level water bodies detection. This script does the binary classification, i.e. water bodies (represented in blue color) and non water bodies (represented in true RGB color composite).

Script Workflow: Simple Water Bodies Mapping Process
The workflow of simple water bodies mapping process is illustrated in Figure 1 below in which the step and corresponding snippet are presented by the same outline color. Before running the script, the study area and source was selected.

The script starts with defining the **source** (S2L2A, S2L1C and L8), histogramIndex and threshold parameters. The **calculatePixel(p)** function is main part of the script. The function starts with defining the bands based on defined data source that include: **r** (red), **g** (green), **b** (blue), **nr** (near infrared), **s1** (short-wave infrared 1), and **vre1** (vegetation red edge 1) bands. Threshold values for indices might/would vary for different study area. By defining, suitable histogramIndex parameter (0: MNDWI, 1: NDWI, 2: SWI) based on used data source and executing the script one by one for the study area, the corresponding histogram can be visualized in histogram chart through histogram function within platform, from which a threshold could be determined for the selected AOI and then a subsequent default threshold parameter could be updated. The threshold value for analyzed index is between left and right peak of the histogram.

The water body detection is carried out considering one or multiple indices, as mentioned below. However, SWI is only applicable for Sentinel dataset as red edge bands are only avalilable in sentinel product.

- MNDWI (Modified Normalized Difference Water Index)
  MNDWI = (Green - SWIR_1) / (Green + SWIR_1)
- NDWI (Normalized Difference Water Index)
  NDWI = (Green - NIR) / (Green + NIR)
- SWI (Surface Water Index)
  SWI = (Vegetation_Red_Edge_1 - SWIR_1) / (Vegetation_Red_Edge_1 + SWIR_1)

The applicable above mentioned indices are calculated for each pixel and compared with the threshold defined initially to detect the water bodies in **wbiS2** function for sentinel data as a data source. In this case, if any of these water indices values exceed the corresponding thresholds, that pixel is classified as water. Sentinel-2 Level2A data provide a scene classification (SCL) map thata consist of cloud scene classes. Thus, cloud scene classes 8 (cloud medium probability) and 9 (cloud high probability) are checked and later filtered out from the output. In color definition block, pixel that is identified as water body is assigned with blue color and the non-water bodies pixels are represented in true RGB color. Additionally, in case of S2 L2A, pixel is displayed in true RGB color if pixel is classified as cloud scene in the SCL map.

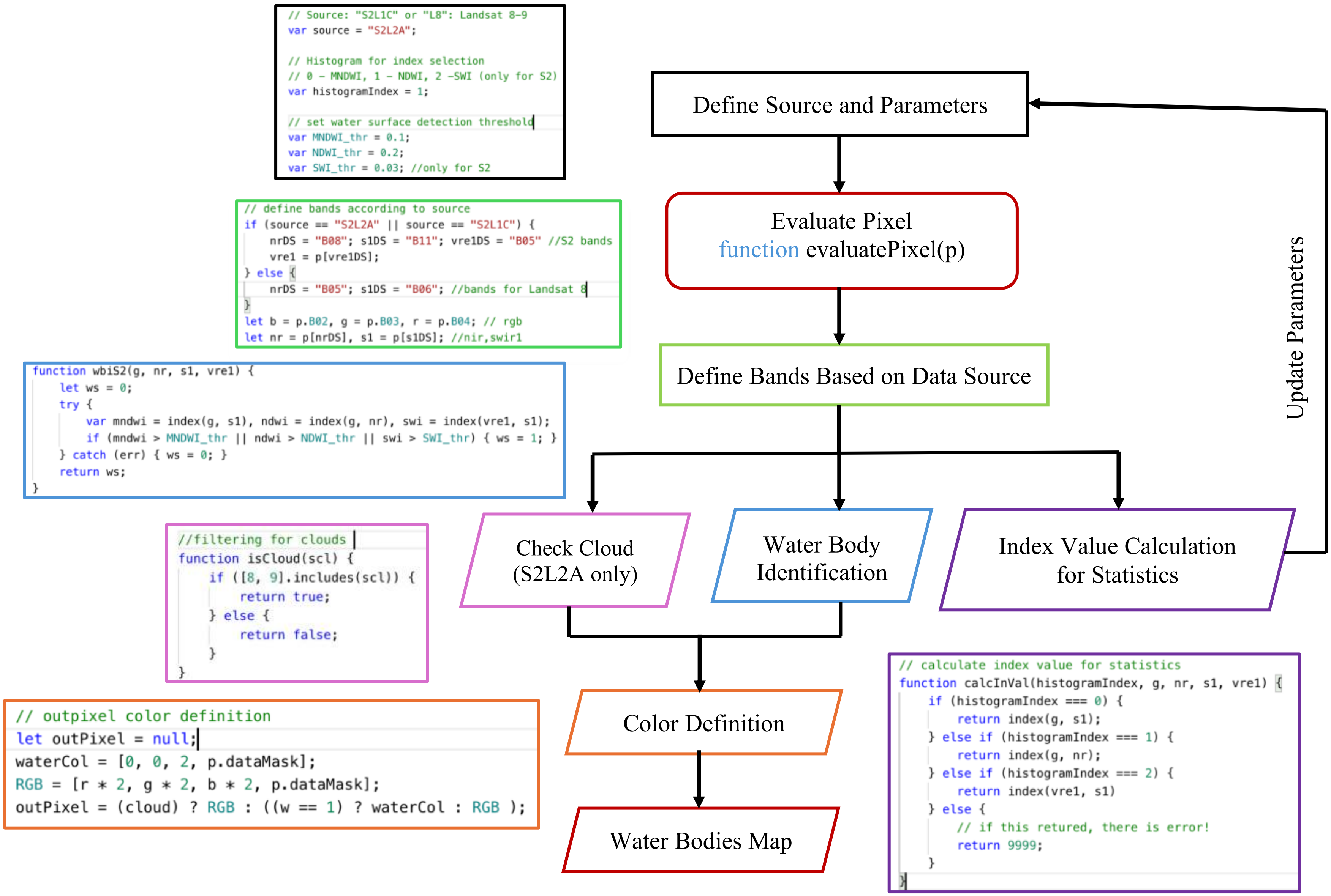

Figure 1: Simple Water Bodies Mapping

Case Study 1 : Prasae Reservoir, Rayong, Thailand

Prasae Reservoir located in Rayong, Thailand was selected as a study area. Cloud free sentinel-2 Level2A data of acquisition date 2024-11-11 was selected for the analysis. Figure 2 represents image of the study area visualized in false color composite.

Figure 2: Prasae Reservoir

Then, the selected script was run to determine the indices threshold values from the histogram diagram. The script was run three times to determine MNDWI, NDWI and SWI threshold and obtained histograms are represented in Figure 3 , Figure 4 and Figure 5 respectively. Based on these histograms, the threshold values for the MNDWI, NDWI, and SWI indices for the Sentinel-2 L1C dataset are 0.01, 0.04, and 0.09, respectively were finalazed.

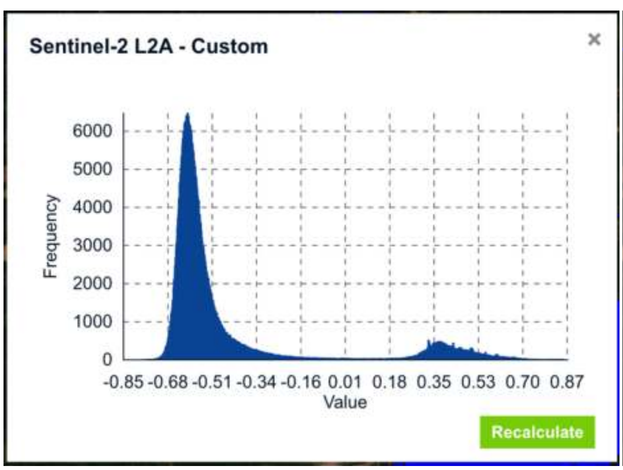

Figure 3: MNDWI Histogram

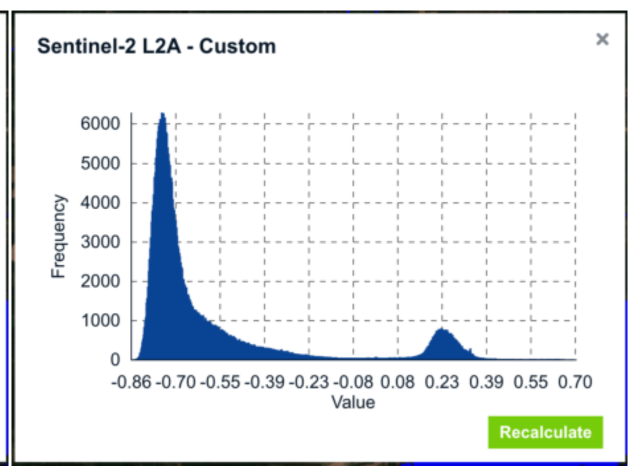

Figure 4: NDWI Histogram

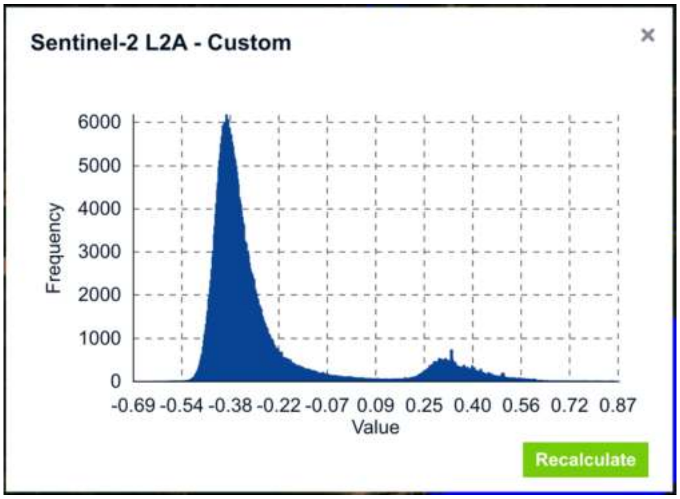

Figure 3: SWI Histogram

The default indices threshold parameters were updated using the value obtained from histogram analysis. Figure 6 represents the water bodies map obtained after running the script with updated parameters.

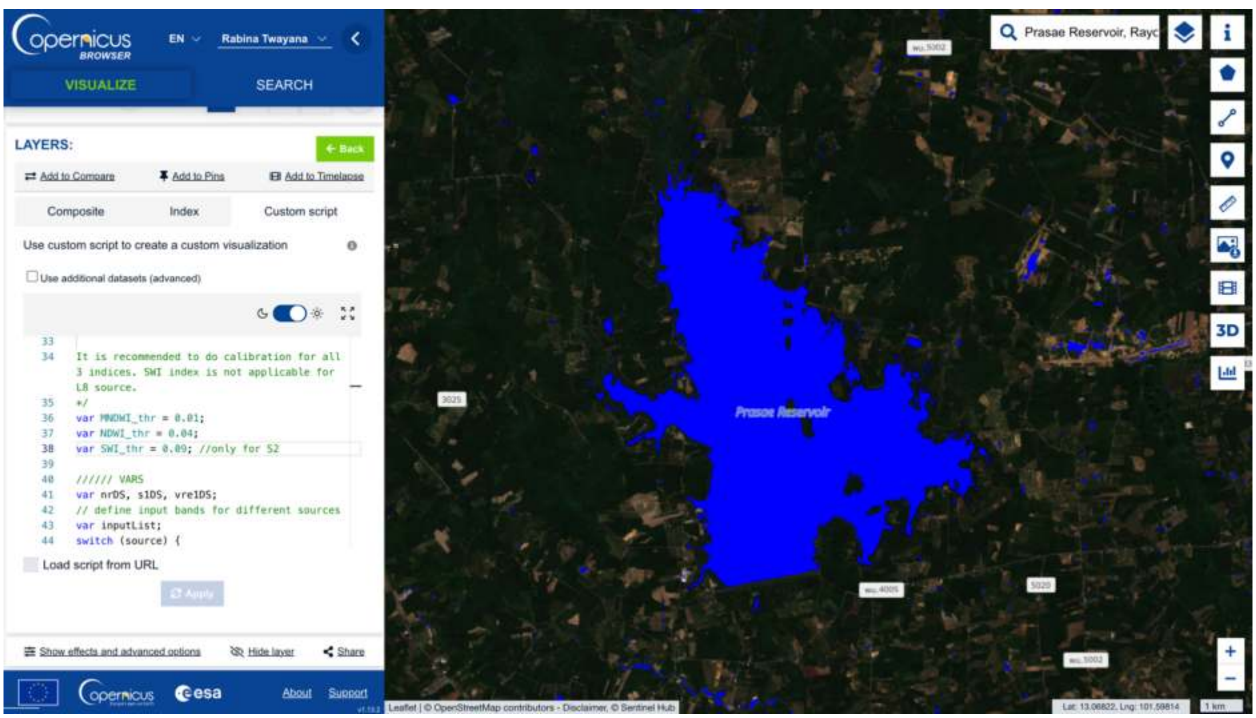

Figure 4: Water Bodies Map

Figure 7 represents the time series map of Prasae reservoir over the past two years focusing on cloud coverage of less than 15%. Minimum cloud coverage images were not found majorly during June to October. Time series analysis for the available months indicates that the water extent was lower between February and April and higher during November and December.

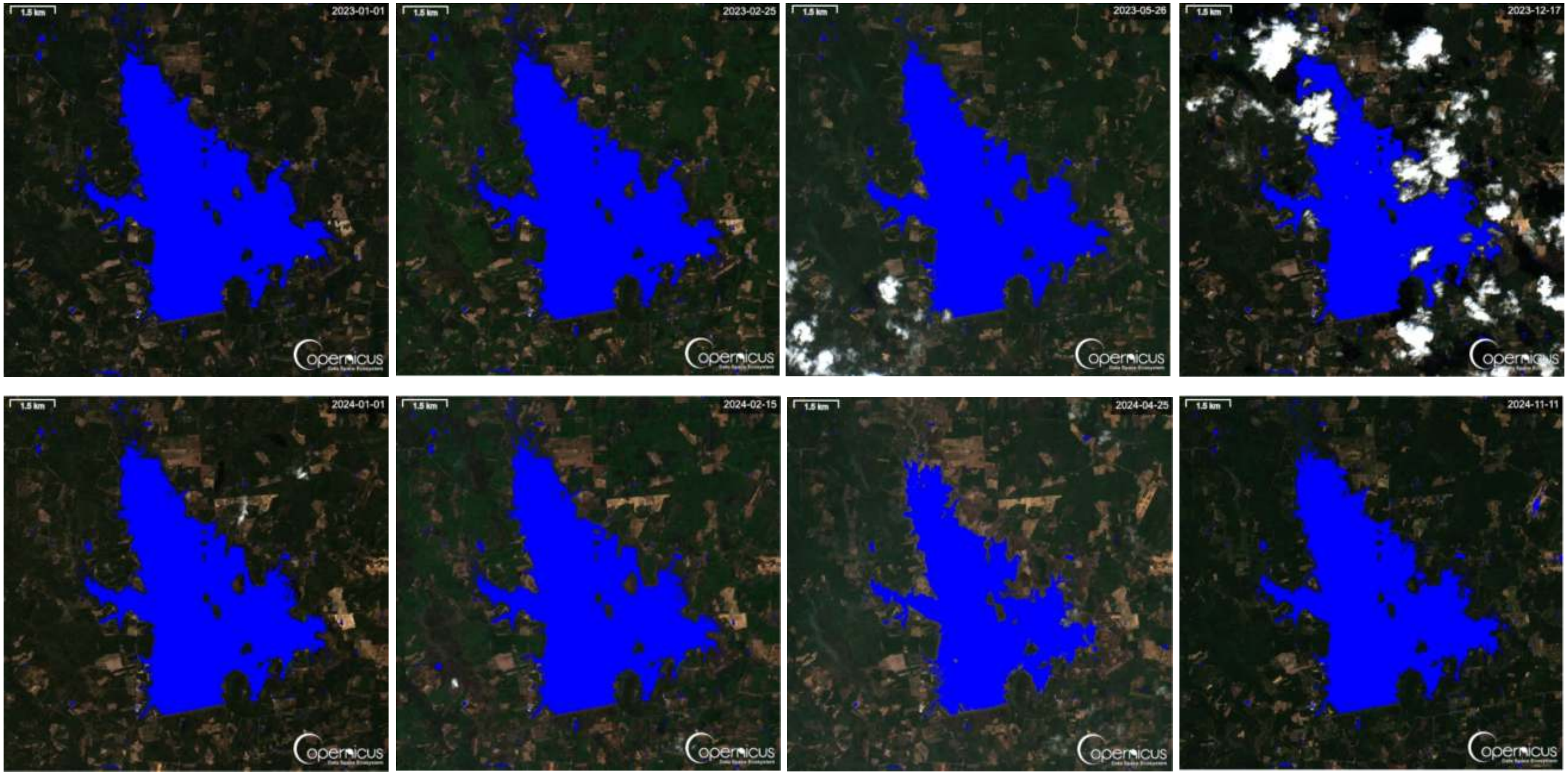

Figure 5: Monthly Timeseries Water Bodies Map (2023-2024)

Case Study 2: Water Bodies Mapping of Salzburg City, Austria using Sentinel-2 L1C and L2A Dataset

The threshold values for water indices will differ when mapping water bodies using Sentinel-2 L1C and L2A, even though applied for the same study area. The analysis was conducted on data acquired on 2024-10-29 that has cloud coverage of 1.5%. The generated histogram is presented in Figure 8 and Figure 9 below. Based on these histograms, the finalized threshold values for the MNDWI, NDWI, and SWI indices for the Sentinel-2 L2A dataset are 0.2, 0.2, and 0.2, respectively. Similarly, for the Sentinel-2 L1C dataset, the selected threshold values for MNDWI, NDWI, and SWI are 0.40, 0.40, and 0.30, respectively.

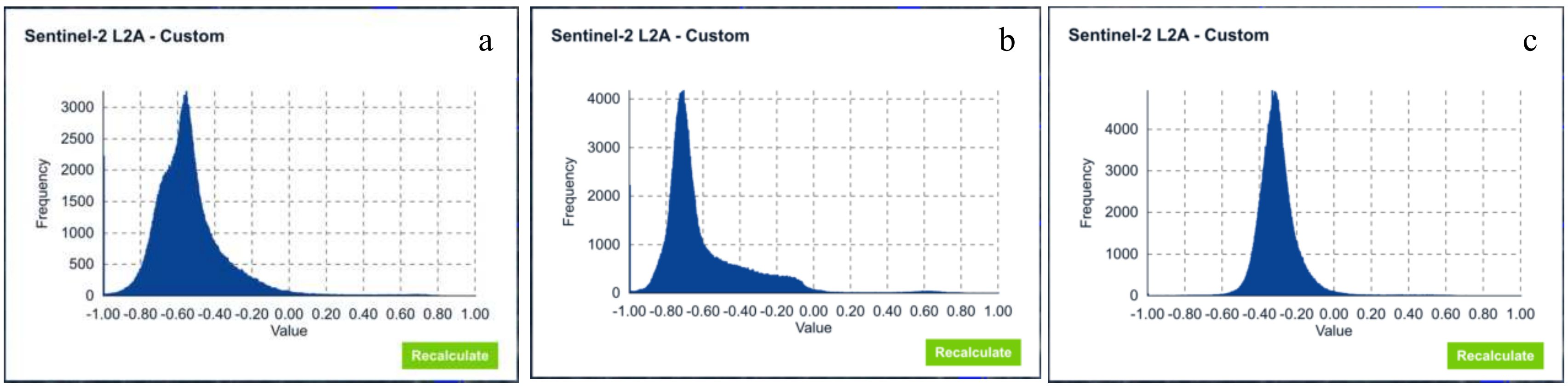

Figure 8: Histogram using Sentinel-2 L2A dataset (a) MNDWI , (b) NDWI, (c) SWI

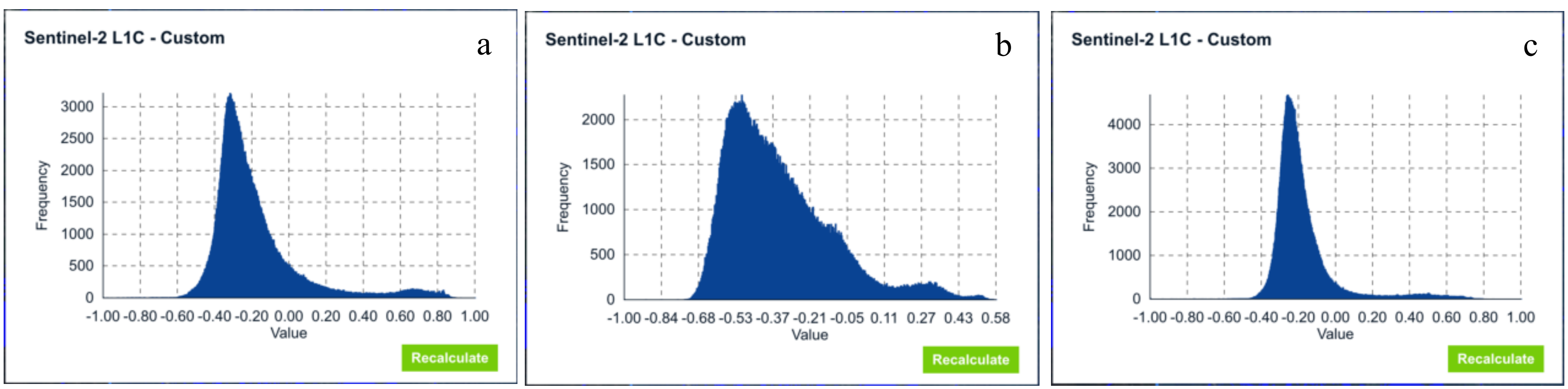

Figure 6: Histogram using Sentinel-2 L1C Dataset (a) MNDWI , (b) NDWI, (c) SWI

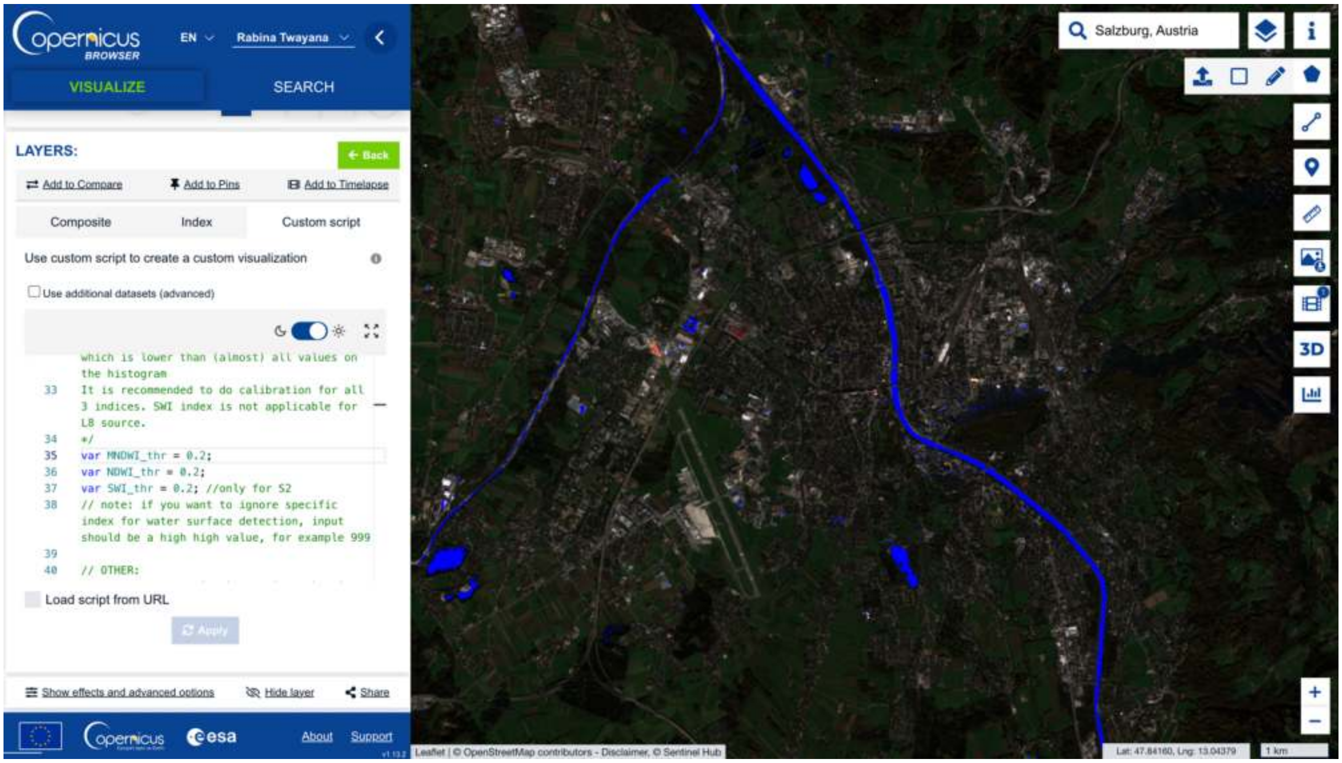

Figure 8: Water Bodies Map using Sentinel-2 L2A data

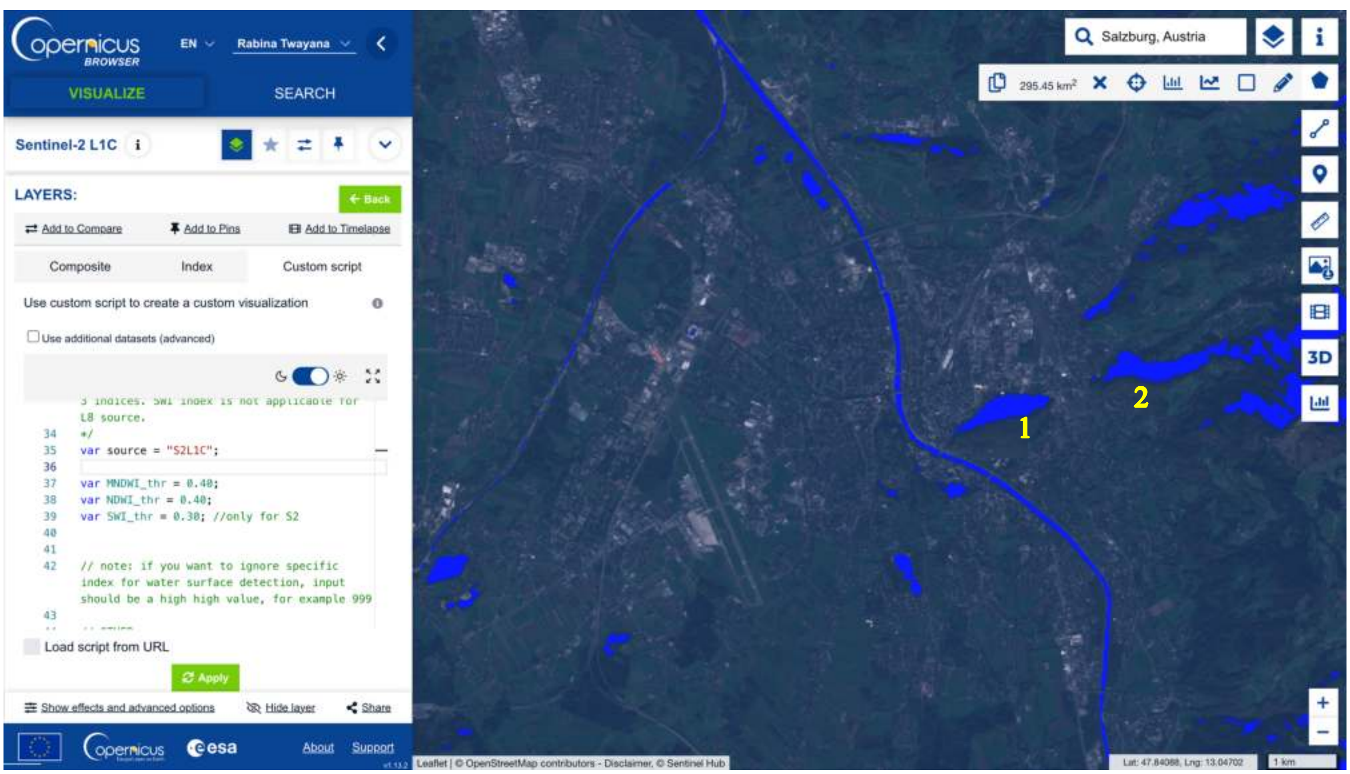

Figure 8: Water Bodies Map of Salzburg City using Sentinel-2 L1C data

Figure 10 and Figure 11 represents the water bodies map of Salzburg city generated using Sentinel-2 L2A and Sentinel-2 L1C data respectively. The major difference can be witnessed in Eastern part of the region (river). The shadowed forest area (for eg: location 1 and 2) marked in Figure 11 and referenced in Figure 12 (Google Earth map) below, were misclassified as an water bodies using Sentinel-2 L1C data as shadowed region also has the less reflectance like water bodies. Thus, shadow and cloud effect need to be consider if analysis is carried out in Sentinel-2 L1C in this case.

Figure 9: Salzburg Region Map (Source: Google Earth)